Department of Electrical and Computer Engineering
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ELEC-271 Digital Systems

Problem Set 7

1. Design a circuit that realizes the FSM defined by the following state-assigned table using *D* flip-flop. Derive the expressions for the next-state variables and output.

Present state y_2y_1	Next state $w = 0$ Y_2Y_1	Next state $w = 1$ Y_2Y_1	Output z
0 0	1 0	1 1	0
0 1	0 1	0 0	0
1 0	1 1	0 0	0
1 1	1 0	0 1	1

2. Repeat Q1 using *JK* flip-flops.

3. Show a state table for the state-assigned table in Q1, using *A*, *B*, *C*, and *D* for the states. Give a new state-assigned table using a one-hot encoding. Synthesize a circuit using *D* flip-flops. You do not need to draw the circuit; just derive the expressions for the next-state variables and output.

4. Derive the state diagram for an FSM that has an input *w* and an output *z*. The machine has to generate $z = 1$ when the previous four values of *w* were 1001 or 1111; otherwise, $z = 0$. Overlapping input patterns are allowed. An example of the desired behaviour is:

w: 010111100110011111
z: 000000100100010011

5. Write a behavioural VHDL code for the FSM described in Q4.

6. Design a minimal state table for a circuit that has an input *w* and an output *z*. The Moore-type FSM produces an output of 1 if in the input sequence it detects either 110 or 101 patterns. Overlapping sequences should be detected.

7. A sequential circuit has two inputs w_1 and w_2, and an output z. Its function is to compare the input sequences on the two inputs. If $w_1 = w_2$ during any four consecutive clock cycles, the circuit produces $z = 1$; otherwise, $z = 0$. For example:

w_1: 0110111000110
w_2: 1110101000111
z: 0000100001110

8. In Figure 8.59 of textbook assume that the unspecified outputs in states B and G are 1 and 0, respectively. Derive the minimized state table for this FSM.

9. Design a counter that counts pulses on line w and displays the count in sequence 0, 2, 1, 3, 0, 2, …. Use D flip-flops in your circuit.

10. Repeat Q9 using JK flip-flops.

11. Design a modulo-6 counter, which counts in the sequence 0, 1, 2, 3, 4, 5, 0, 1, …. The counter counts the clock pulses if its enable input, w, is equal to 1. Use T flip-flops in your circuit.

12. Write a behavioural VHDL code for the FSM shown in Figure 8.58 of textbook, using the style of code in Figure 8.33.